Why Did American Student Achievement Decline?

Verifying the Trends and Testing Competing Explanations with NAEP Data

June 2026

Abstract

American student achievement has fallen substantially over the past decade. Using data drawn directly from the National Assessment of Educational Progress (NAEP) Data Service API—national means and percentiles for mathematics and reading in grades 4 and 8 (1990–2024), the Long-Term Trend (LTT) assessment (1971–2023), state-level means, and demographic subgroups—this report first verifies the decline and characterizes its timing, distribution, and geography, then systematically evaluates eight candidate explanations. The data show a *two-phase* decline. Phase one (roughly 2013–2019) was a slow erosion concentrated almost entirely among the lowest-performing students: at age 13, scores at the 10th percentile of the LTT mathematics distribution fell 12.6 points between 2012 and 2020 while the 90th percentile was flat. Phase two (2020–2022) was an abrupt, across-the-board drop coinciding with pandemic schooling disruptions, followed by a weak and incomplete recovery that has stalled at the bottom of the distribution. By 2024, grade 8 reading stood at its lowest level ever recorded and the 90th–10th percentile gap in every grade–subject combination was the widest on record. Testing hypotheses against the timing, distributional, subgroup, geographic, and international evidence: pandemic disruption clearly caused phase two but cannot explain phase one; chronic absenteeism roughly doubled after 2020 and is a leading brake on recovery; demographic change, school funding cuts, and teacher shortages contribute little. Three further discriminating tests sharpen the verdict on phase one. A cohort-matched decomposition shows that pre-pandemic grade 8 losses arose *during* the middle-school years—cohorts arrived at grade 4 at record levels and lost ground between grades 4 and 8—identifying a period effect concentrated in adolescence. Catholic schools, never subject to federal test-based accountability, declined in step with public schools before the pandemic, and an event-study analysis of states' staggered NCLB-waiver adoption finds no effect of accountability relaxation on bottom-decile scores. Together these results weaken the accountability-retreat hypothesis and leave the post-2012 saturation of adolescent life by smartphones and digital media—which uniquely predicts an adolescent-specific, sector-neutral, geographically uniform, and international decline—as the explanation most consistent with the full pattern of evidence, compounded by a pandemic shock whose effects have not been remediated. Two mechanism checks complete the picture: between-state variation in 2020–21 virtual schooling explains almost none of the between-state variation in NAEP declines (the closure effect, well identified at the district level, washes out under state aggregation), while NAEP's own attendance item shows student-reported absence rising from 2015–2019—before the pandemic—and a Kitagawa decomposition attributes 25–42% of the 2019–2024 declines, but only 7–19% of the pre-pandemic grade 8 declines, to the shift of students into high-absence categories.

1 Introduction

For two decades beginning in the early 1990s, American students made steady, historically large gains on the National Assessment of Educational Progress (NAEP), the nation's longest-running and most credible barometer of K–12 achievement. Fourth-grade mathematics scores rose nearly a full standard deviation between 1990 and 2013. That era is over. Every major NAEP series peaked between 2012 and 2015 and has fallen since, with the steepest losses after 2019. The reversal is large: by 2024, average grade 8 mathematics scores had returned to their 2000 level, and grade 8 reading scores were the lowest ever recorded in the assessment's history.

This report does two things. First, it *verifies* the decline directly from primary data, pulling national, state, percentile, and subgroup statistics from the NCES NAEP Data Service API rather than relying on secondary summaries (National Center for Education Statistics, 2025). Second, it *tests* eight candidate explanations against five kinds of evidence: (i) the timing of the decline; (ii) its distribution across the achievement spectrum; (iii) its incidence across demographic subgroups; (iv) its geography across states; and (v) its presence or absence in other countries. Most public discussion attributes the decline to the COVID-19 pandemic. A central finding of this report is that the pandemic explanation, while clearly correct for the 2019–2022 drop, is incomplete: between a quarter and three-quarters of the total decline (depending on grade and subject) either predates March 2020 or postdates the return to in-person schooling, and the pre-pandemic phase has a distinctive signature—losses concentrated almost entirely among low performers—that any complete explanation must fit.

The analysis is descriptive and abductive rather than causal in the experimental sense: national test-score trends do not admit randomized variation, and several candidate causes operate simultaneously. The discipline imposed here is that each hypothesis generates predictions about timing, distribution, subgroups, geography, and international comparability, and is scored against all five. Hypotheses that fit one fact but contradict another are weighed accordingly.

2 Data

Main NAEP. The primary data are national public- and private-school average scale scores and percentile scores (10th, 25th, 50th, 75th, 90th) for mathematics and reading at grades 4 and 8, for all assessment years from 1990 (mathematics) and 1992 (reading) through 2024, retrieved from the NAEP Data Service API (National Center for Education Statistics, 2025). Subgroup means (race/ethnicity, sex, National School Lunch Program eligibility), school-sector results (public, Catholic, private), state-level means for 2013–2024, and a state $\times$ year panel of percentile scores (2003–2024) were retrieved from the same source; state ESEA-waiver approval dates were compiled from Department of Education records, CRS Report R42328, and EdWeek's state-by-state tracking.[1] NAEP scale scores are reported on 0–500 scales; the within-grade student standard deviation (SD), estimated from the 2013 percentile spread, is roughly 29 points for grade 4 mathematics, 36 for grade 8 mathematics, and 34–36 for reading. A useful rule of thumb is that 10–12 NAEP points correspond to roughly one grade level of learning.

[1] NSLP-eligibility breakdowns are available through 2022; the API does not return them for 2024. Pre-2003 national figures use the no-accommodations samples for 1990–1996 and accommodated samples thereafter, following NAEP reporting conventions. All scores were re-validated against published NAEP report-card values.

Table 1: National average score changes by period (NAEP scale points).

Series	Change by period			Total 2013–2024	
	2013–19	2019–22	2022–24	Points	SD units
Mathematics, grade 4	−1.0	−4.6	+1.5	−4.1	−0.14
Mathematics, grade 8	−2.6	−7.7	−0.4	−10.8	−0.30
Reading, grade 4	−1.4	−3.0	−2.5	−6.8	−0.19
Reading, grade 8	−4.4	−2.7	−2.4	−9.5	−0.28

Long-Term Trend (LTT) NAEP. The LTT assessment has measured 9-, 13-, and 17-year-olds with substantially unchanged content since the early 1970s, providing the longest consistent yardstick. National means and percentiles for ages 9 and 13 (1971–2023) were compiled from the Digest of Education Statistics and the LTT Data Service and cross-validated against published highlights (National Center for Education Statistics, 2023). Two timing facts make the LTT unusually valuable here: the age 9 “2020” assessment was administered in January–March 2020, *immediately before* the pandemic school closures, and the age 13 “2020” assessment in fall 2019. These provide clean pre-pandemic endpoints.

International assessments. U.S. and OECD-average PISA scores (2000–2022) and U.S. TIMSS mathematics scores (2011–2023) are taken from NCES and OECD publications (OECD, 2023; National Center for Education Statistics, 2024).

Contextual evidence. Published causal and descriptive research is used to test specific hypotheses: pandemic schooling-mode studies (Goldhaber et al., 2023; Jack et al., 2023), district-level recovery analyses (Education Recovery Scorecard, 2024, 2025), absenteeism tracking (Malkus, 2025; Dee, 2024), accountability research (Dee and Jacob, 2011), and school-finance research (Jackson et al., 2016; Jackson and Mackevicius, 2024; Jackson et al., 2021).

3 Verifying the Decline

3.1 National trends: a peak around 2013, then two distinct drops

Figure 1 plots national average scale scores for the four main NAEP series. All four rose strongly through the 2000s, peaked in 2013 (grade 4 reading in 2015, fractionally above its 2013 level), drifted downward through 2019, dropped sharply between 2019 and 2022, and—except for a partial rebound in grade 4 mathematics—continued falling or stagnated through 2024.

Table 1 decomposes the decline. Three observations matter for everything that follows. First, the total losses are large: 0.14–0.30 student-level SD, roughly one grade level of learning in grade 8. In level terms, 2024 grade 4 mathematics matches 2005; grade 8 mathematics matches 2000; grade 4 reading is below every year since 1998; and grade 8 reading (258.0) is below its previous historical minimum. Second, the pandemic window accounts for the majority of the mathematics decline (71% of the grade 8 total) but *less than half* of the reading decline at grade 8, where 46% of the loss had already occurred by 2019. Third, the post-pandemic period has produced essentially no aggregate recovery: only grade 4 mathematics regained ground (about a third of its pandemic loss), while both reading series fell *further* between 2022 and 2024.

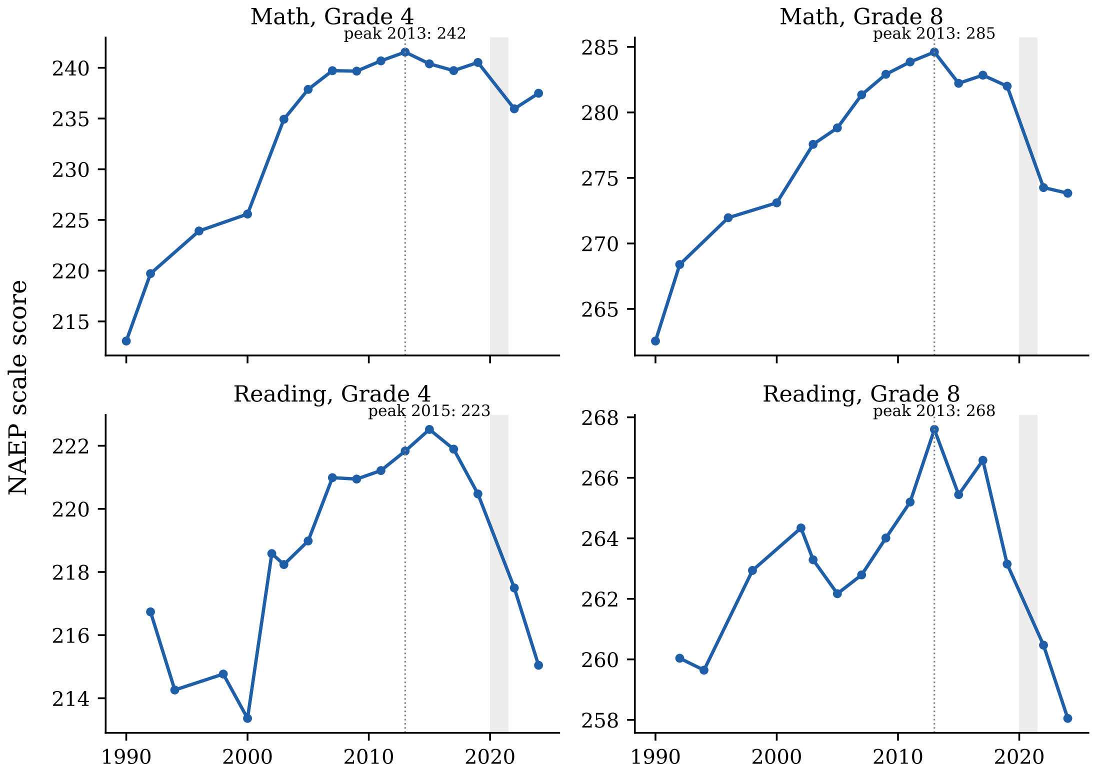

Figure 1: National average NAEP scale scores, 1990–2024. Dotted line marks 2013; shaded band marks the period of pandemic-disrupted schooling. Data: NAEP Data Service API.

3.2 The Long-Term Trend assessment confirms the timing

The LTT data (Figure 2) rule out the possibility that the main-NAEP decline is an artifact of framework or administration changes. On test content held essentially constant since the 1970s, age 9 and age 13 scores peaked in 2012 and were already falling before the pandemic: age 13 mathematics fell from 285 in 2012 to 280 in fall 2019, and age 13 reading from 263 to 260, *before any school closed*. The pandemic then produced the largest drops ever recorded in the series: age 9 mathematics fell 7 points between early 2020 and 2022 (its first statistically significant decline ever), and age 13 mathematics fell 9 further points by fall 2022, landing at its 1992 level. Age 13 reading in 2023 (256) was statistically indistinguishable from its 1971 baseline—a half-century of progress erased for the median young adolescent.

3.3 The distributional signature: collapse at the bottom

The single most diagnostic fact in the data is *where* in the achievement distribution the losses occurred. Figure 3 plots score changes since 2013 at the 10th through 90th percentiles for grade 8; Figure 4 shows the same comparison for the LTT, split into the pre-pandemic (2012→2020) and pandemic (2020→2022/23) windows.

Before the pandemic, the decline was almost exclusively a bottom-of-the-distribution phenomenon. Between 2012 and 2020, age 13 LTT mathematics fell 12.6 points at the 10th percentile, 7.4 at the 25th, 4.4 at the median—and *rose* 0.1 points at the 90th. Main NAEP shows

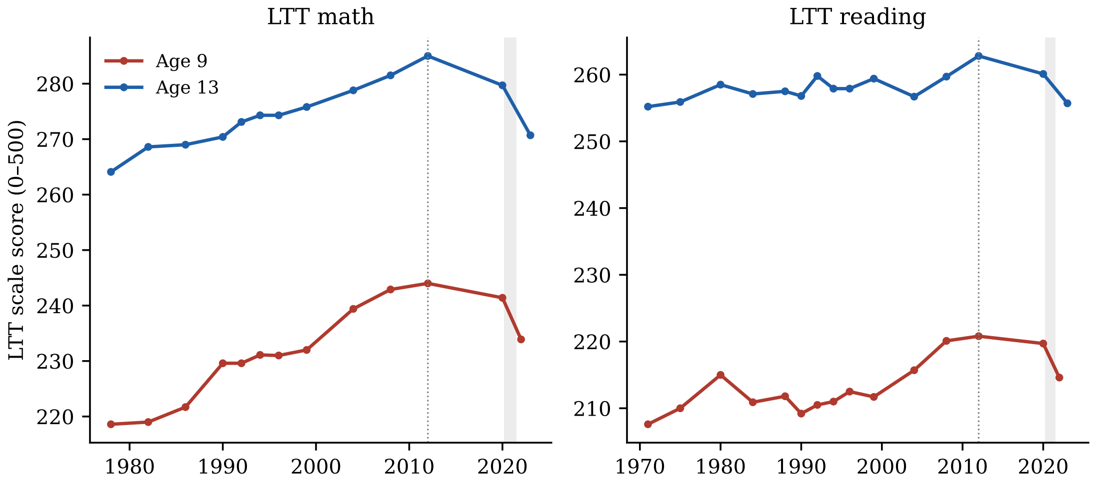

Figure 2: NAEP Long-Term Trend average scores, ages 9 and 13, 1971–2023. The age 9 "2020" point was assessed January–March 2020 (pre-pandemic); the age 13 "2020" point in fall 2019. Dotted line marks 2012. Data: NCES Digest tables 221.85/222.85, validated against the LTT Data Service.

the identical pattern: between 2013 and 2019, grade 8 mathematics fell 6.6 points at the 10th percentile while *gaining* 2.7 points at the 90th; grade 8 reading fell 10.2 points at the 10th percentile against 1.1 at the 90th. America's strongest students were holding steady or improving through 2019; its weakest students were in free fall years before anyone had heard of COVID-19.

The pandemic broke this pattern in an informative way: 2019–2022 losses appear across the entire distribution—in grade 8 mathematics the drop was essentially uniform, 6–9 points at every percentile, consistent with a shock (school disruption) that hit all students rather than only struggling ones. After 2022, the top of the distribution began recovering while the bottom kept falling: in grade 4 mathematics, 2022–2024 changes ranged from +2.7 points at the 75th percentile to −0.6 at the 10th.

The cumulative effect on inequality is unprecedented in NAEP's history. The 90–10 gap (Figure 5) widened between 2013 and 2024 by 13 points in grade 4 mathematics (75→89), 16 points in grade 8 mathematics (93→109), 15 points in grade 4 reading, and 16 points in grade 8 reading—in every case the widest gap ever recorded.

3.4 Subgroups: declines within every group

Score declines appear *within* essentially every demographic subgroup, which constrains compositional explanations (Section 5.5). Between 2013 and 2024, grade 8 mathematics fell 8.3 points among White students, 11.4 among Black students, 13.1 among Hispanic students, and was about flat (−1.1) only among Asian/Pacific Islander students. Both NSLP-eligible (lower-income) and non-eligible students lost ground through 2022 (grade 8 mathematics: −10.4 and −9.9 respectively). Pre-pandemic, the within-group pattern mirrors the percentile pattern: between 2013 and 2019 grade 8 mathematics scores among lunch-eligible students fell 3.6 points while non-eligible students lost only 1.0; Asian/Pacific Islander students *gained* 3.5 points. Figure 6 shows the level trends by lunch eligibility.

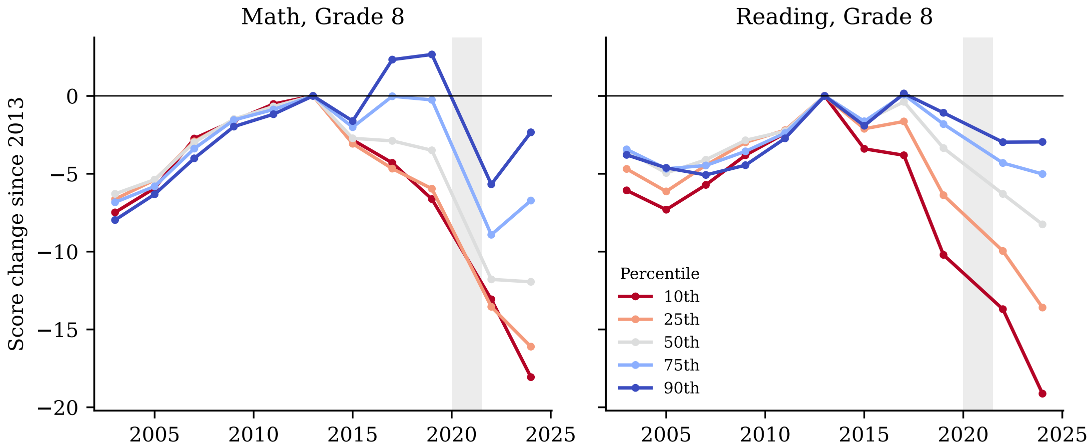

Figure 3: Grade 8 score changes since 2013 by percentile. Data: NAEP Data Service API.

3.5 Geography: a national phenomenon, modulated by pandemic policy

The decline is not regional. Between 2013 and 2019, 75–94% of states (depending on series) posted declines; by 2024, between 88% and 100% of states remained below their 2013 level. No state escaped the national trend in grade 8 mathematics. The two prominent partial exceptions are instructive: Mississippi rose from 49th to 29th in grade 4 reading between 2013 and 2019 following its 2013 early-literacy reform (Spencer, 2024), and a handful of states (e.g., Alabama, Louisiana) regained or exceeded pre-pandemic grade 4 mathematics levels by 2024.

State variation *within* the pandemic window lines up with schooling-mode differences (Section 5.1). States' 2019–2022 drops in grade 4 mathematics ranged from about -13 points to roughly zero. The states that fell most recovered somewhat faster afterward (correlation between 2019–22 change and 2022–24 change ≈ -0.3 to -0.5; Figure 7), consistent with partial mean reversion as in-person schooling resumed—but the rebound slopes are shallow (about -0.4), implying that most of the pandemic-window loss has, so far, been persistent rather than transitory.

3.6 International context: America is not alone

The U.S. decline is part of a broader rich-world phenomenon (Figure 8). The OECD-average PISA mathematics score fell 22 points between 2012 and 2022 (494→472) and reading fell 20 points—declines that began *before* the pandemic. The OECD itself concluded that the fall was "only partly attributable to the COVID-19 pandemic," noting that reading and science had been sliding for roughly a decade (OECD, 2023). Twelve OECD systems—including Finland, the Netherlands, Belgium, and Canada—show statistically significant mathematics declines beginning before 2018. Notably, U.S. PISA *reading* was flat (505→504) between 2018 and 2022 while the OECD average fell 11 points, so the U.S. position relative to peers actually improved even as its NAEP reading scores fell; PISA samples 15-year-olds, whose NAEP-cohort losses were smaller than younger students'. U.S. TIMSS mathematics tells the same story as NAEP: grade 8 scores fell 27 points between 2019 and 2023, back to mid-1990s levels, with the lowest 10th-percentile scores since the study began (National Center for Education Statistics, 2024).

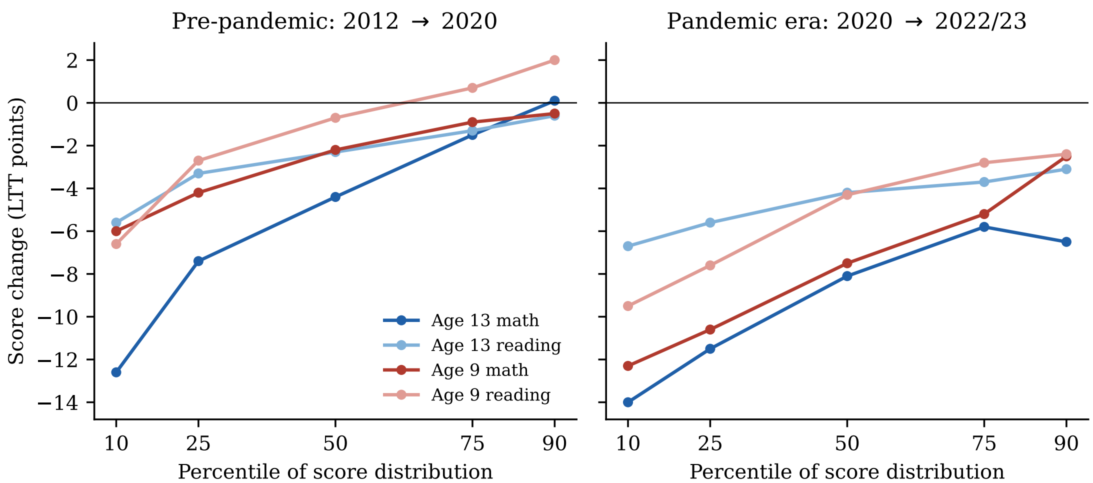

Figure 4: LTT score changes by percentile: pre-pandemic (left) versus pandemic era (right). The pre-pandemic decline is concentrated almost entirely among low performers; the pandemic-era decline hits everywhere but remains steepest at the bottom. Data: LTT Data Service.

3.7 Summary of facts to be explained

Any successful explanation must account for: (F1) a peak in 2012–2013 and slow decline through 2019; (F2) the pre-2019 losses concentrated overwhelmingly at the bottom of the distribution, with the top flat or rising; (F3) a large, across-the-distribution drop in 2019–2022; (F4) continued post-2022 decline in reading and at the bottom of every distribution, against partial recovery at the top and in younger students' mathematics; (F5) declines within every demographic group and nearly every state; (F6) similar pre-pandemic declines across many rich countries; and (F7) record-high achievement inequality.

4 Candidate Explanations

Eight hypotheses recur in research and public debate:

H1. **Pandemic disruption.** School closures, remote instruction, and associated social disruption caused learning loss.

H2. **Smartphones and digital media.** The post-2012 saturation of adolescent life by smartphones and social media displaced reading, homework, and attention.

H3. **Accountability retreat.** The dismantling of No Child Left Behind's test-based accountability (waivers from 2012, ESSA from 2015) removed pressure that had propped up achievement among low performers.

H4. **Great Recession funding cuts.** State school-funding cuts after 2008 reduced school quality for the cohorts tested in the mid-2010s.

H5. **Demographic change.** A rising share of disadvantaged students mechanically lowered average scores.

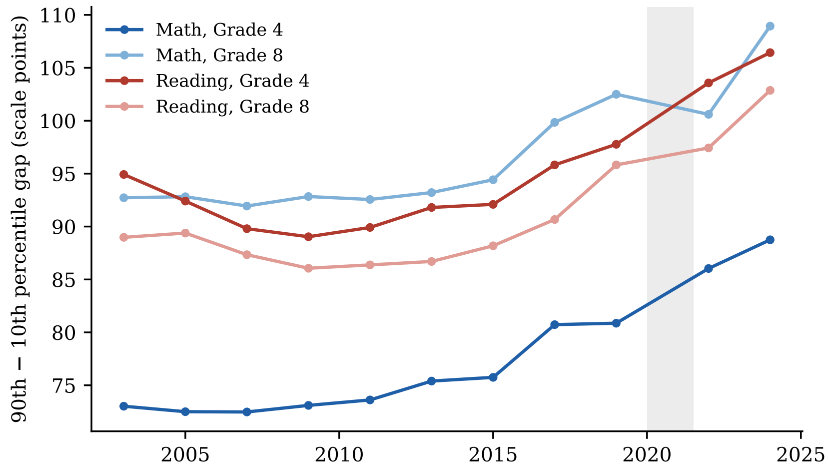

Figure 5: Gap between the 90th and 10th percentile scores, 2003–2024. Data: NAEP Data Service API.

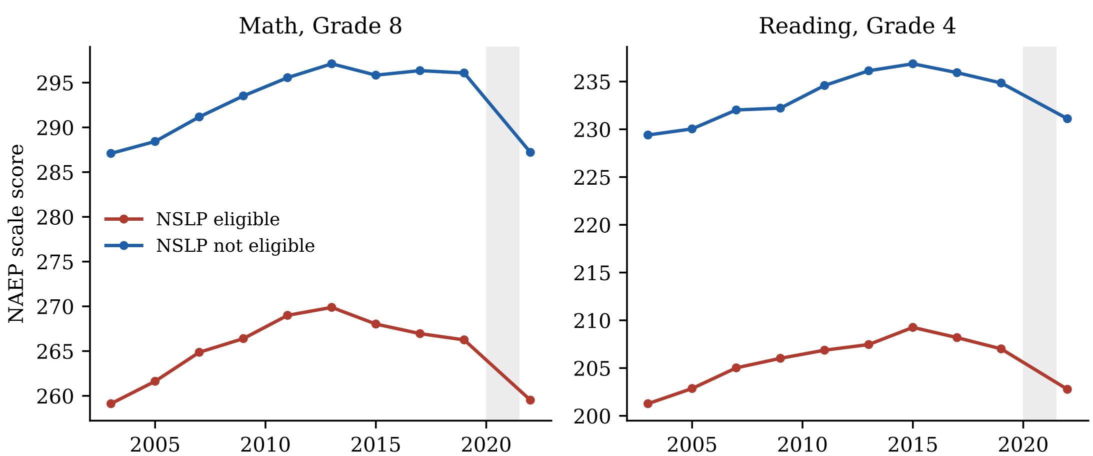

Figure 6: Average scores by National School Lunch Program eligibility (available through 2022). Data: NAEP Data Service API.

H6. **Chronic absenteeism.** Students stopped attending school regularly after the pandemic.

H7. **Declining reading practice.** Children read less, for school and pleasure, eroding literacy skills.

H8. **Other: teacher shortages, lowered expectations/grade inflation.**

5 Testing the Hypotheses

5.1 H1: Pandemic disruption — decisive for 2019–2022, irrelevant before, incomplete after

Predictions. A sharp drop between the 2019 and 2022 assessments; larger losses where schooling was remote longer; recovery as schooling normalized; no effect before 2020.

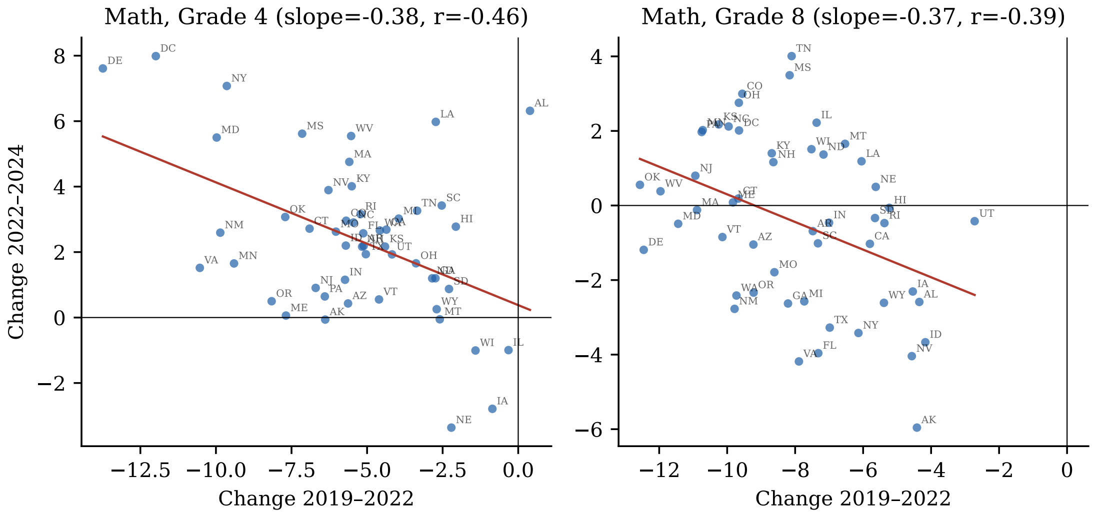

Figure 7: State-level pandemic losses versus post-pandemic recovery, mathematics. Each point is a state. Data: NAEP Data Service API.

Evidence. The 2019–2022 drop is unambiguous and its causal link to schooling disruption is among the best-documented findings in modern education research. Districts that were remote for more than half of 2020–21 saw mathematics achievement growth shortfalls of 0.44 SD in high-poverty schools, versus about 0.17 SD where instruction stayed in person (Goldhaber et al., 2023). Across twelve states, district math pass rates fell 14.2 percentage points on average in spring 2021, but only 4.1 points in districts that remained fully in person (Jack et al., 2023). The dose-response gradient, its replication across data sets, and the coincidence in time make H1 conclusively established for phase two—particularly for mathematics, which is learned mostly in school. Consistent with this, the 2019–2022 NAEP drop was nearly uniform across the achievement distribution (Section 3.3), as expected from a shock to schooling itself.

Two equally important negative results bound the hypothesis. First, H1 predicts nothing before 2020, yet roughly a quarter to a half of the total 2013–2024 decline predates the pandemic (Table 1), including the entire 2012–2020 collapse at the bottom of the LTT distribution. Second, H1 predicts recovery, and recovery has been feeble: by spring 2023 students had regained only about a third of pandemic mathematics losses and a quarter of reading losses (Education Recovery Scorecard, 2024); by 2025 only 17% of students were in districts at or above 2019 mathematics levels (Education Recovery Scorecard, 2025); and NAEP reading fell *further* after 2022. The ~ -0.4 slope in Figure 7 says the same thing in the NAEP data: states recovered only a fraction of what they uniquely lost.

Verdict. *Confirmed as the dominant cause of phase two (roughly half to three-quarters of the mathematics decline, less of reading); cannot explain phase one; the failure to recover requires additional explanation (H6, and the bottom-distribution forces of H2/H3).*

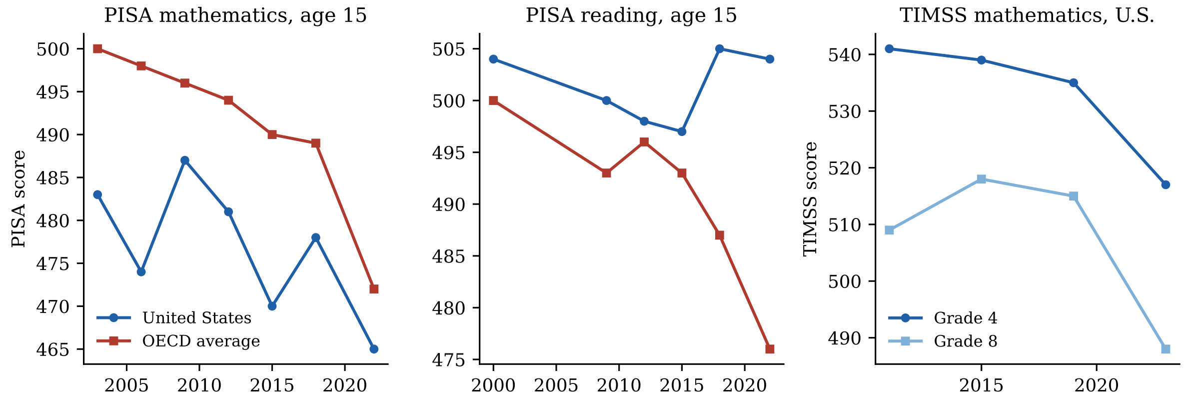

Figure 8: International assessments: PISA mathematics and reading (U.S. vs. OECD average) and U.S. TIMSS mathematics. Data: NCES/OECD publications.

5.2 H2: Smartphones and digital media — the best fit to the international, pre-pandemic facts

Predictions. Decline beginning when smartphones saturated childhood (∼2012–2015); present across countries, subjects, and demographic groups; strongest in reading and among students with the least adult structure; displacement of reading and homework time.

Evidence on timing. Teen smartphone ownership in the U.S. went from 23% in 2011 to 37% in 2012, 73% by 2015, and 95% by 2018, with 45% of teens online "almost constantly" by 2018 (Madden et al., 2013; Anderson and Jiang, 2018). The achievement peak (2012–2013) coincides almost exactly with the takeoff. Adolescent entertainment screen time reached 7h22m/day by 2019 and 8h39m by 2021 (Rideout et al., 2022). The collapse in recreational reading is dramatic and—critically—happened mostly *before* the pandemic: the share of 13-year-olds reading for fun almost daily fell from 27% in 2012 to 17% in 2020 and 14% in 2023, while the share who never or hardly ever read rose from 22% to 31% (Figure 9).

Evidence on breadth. H2 is the only hypothesis that naturally explains fact F6, the synchronized pre-pandemic decline across rich countries with very different school policies, funding trends, and accountability regimes (OECD, 2023). Within PISA 2022, 65% of students across the OECD (66% in the U.S.) reported digital-device distraction in mathematics class; distracted students scored 15 points lower, and students spending 5–7 hours per day on leisure device use scored 49 points below light users after adjusting for socioeconomic status (OECD, 2024). These are associations, not causal estimates, and the OECD says so; but the timing and cross-national synchrony carry the causal weight here.

Fit to the distributional signature. A priori, screen exposure is near-universal, which sits awkwardly with the bottom-concentrated phase-one decline—unless effects are heterogeneous. That is what the broader literature suggests: displacement and distraction harm students with the least self-regulation and the least adult monitoring most (Haidt, 2024), and the reading-for-fun decline was steepest among already-infrequent readers. The hypothesis is thus *consistent* with F2, though it does not uniquely predict it.

Verdict. *Strongly supported as a major driver of phase one (and of the continued post-2022 slide in reading), on timing, international synchrony, and mechanism evidence; the individual-level causal evidence remains associational, so the magnitude is uncertain. The discriminating*

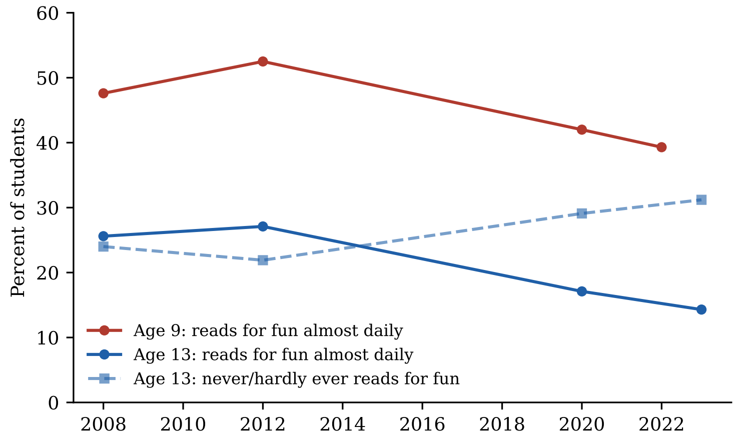

Figure 9: Share of students reading for fun, NAEP LTT student survey. The decline pre-dates the pandemic. Data: LTT Data Service (variable S003501).

tests of Section 6—an adolescent-specific period effect, sector-neutral declines, and the failure of policy-variation tests—all land on the side this hypothesis predicts.

5.3 H3: Accountability retreat — the best fit to the bottom-concentrated U.S. signature

Predictions. Decline beginning after 2011–2012 (NCLB waivers) and deepening after 2015 (ESSA); concentrated among exactly the students accountability targeted—low performers and disadvantaged subgroups; U.S.-specific.

Evidence. The timing matches precisely: the first waivers releasing states from NCLB's consequences were granted in February 2012, covering 34 jurisdictions within a year; the Every Student Succeeds Act (December 2015) made the rollback permanent. The best causal evidence on NCLB found it had raised grade 4 mathematics by roughly 0.23 SD, with gains *concentrated among low-achieving, Black, Hispanic, and lunch-eligible students*, and no effect on grade 4 reading (Dee and Jacob, 2011). The post-2012 decline is nearly a mirror image of those gains: largest at the 10th percentile, among disadvantaged groups, and (initially) clearer in mathematics at the grades accountability touched most. If a policy raised the floor, removing it should drop the floor—and the floor is what dropped (F2). Commentators including Chad Aldeman and former IES director Mark Schneider have pressed this reading of the NAEP record (Aldeman, 2025).

The hypothesis has two limits. It cannot explain F6—Finland and the Netherlands did not pass ESSA—so it cannot be the sole driver. And because accountability operated through schools, it predicts little about the collapse in out-of-school reading (Figure 9). The Mississippi counter-example cuts in H3's favor: the one state that *intensified* early-grade stakes and instructional oversight after 2013 moved sharply against the national trend, with quasi-experimental evidence supporting a genuine policy effect (Spencer, 2024).

Verdict. *On the evidence of this section: supported as a contributor to the U.S. phase-one decline, especially its bottom-concentration; insufficient alone because the decline is international. Section 6 subjects this hypothesis to two direct tests—sector comparison and waiver-timing variation—and both come back unfavorable, substantially weakening it.*

5.4 H4: Great Recession funding cuts — timing partially fits, magnitude does not

Predictions. Declines among cohorts schooled during 2009–2015 austerity, reversing as spending recovered after 2013.

Evidence. Real per-pupil spending bottomed in 2012–13 and 29 states cut per-pupil funding between 2008 and 2016 (Leachman et al., 2017); recession-induced cuts demonstrably reduced achievement (Jackson et al., 2021). But the magnitudes are far too small: the consensus causal estimate is roughly 0.03 SD per $1,000 per pupil sustained for four years (Jackson and Mackevicius, 2024). Recession-era cuts on the order of $1,000–$1,500 per pupil could account for perhaps 0.03–0.05 SD—a fraction of the 0.14–0.30 SD decline—and spending recovered steadily after 2013 while scores kept falling, the opposite of the predicted reversal. Funding also cannot explain F2's top-stability (cuts hit whole districts), F6, or the post-2020 dynamics, when an unprecedented $190 billion federal infusion accompanied historic losses (its measured effect, $\sim$0.008 SD per $1,000 per student, was positive but small relative to the shock).

Verdict. *Rejected as a major cause; plausibly a minor contributor to mid-2010s losses in the states that cut deepest.*

5.5 H5: Demographic change — arithmetic rules it out

Predictions. Stable or rising scores within each subgroup, with composition shifts toward lower-scoring groups driving the aggregate down.

Evidence. The within-group facts in Section 3.4 directly contradict the prediction: scores fell within every major racial/ethnic group, within both lunch-eligibility groups, and for both sexes. Composition shifts were also too slow (Hispanic share up five percentage points over the decade; English learners up about one) to move national means more than a point or two, and the direction of some shifts (record-low child poverty in 2021) runs opposite the score trend. A back-of-envelope reweighting of 2024 grade 8 mathematics subgroup means at 2013 race shares recovers less than 2 of the 10.8 lost points.

Verdict. *Rejected as an explanation of the decline (a 1–2 point compositional drag at most); within-group declines are the phenomenon itself.*

5.6 H6: Chronic absenteeism — a phase-two amplifier and the leading brake on recovery

Predictions. Score declines tracking the rise in absenteeism; weakest recovery where absenteeism stays elevated; no role before 2020.

Evidence. Chronic absenteeism was flat at roughly 15% from 2015 to 2019—so it cannot explain phase one—then nearly doubled to about 28.5% in 2021–22, retreating only to 23.5% by 2023–24 (Malkus, 2025). The persistence of elevated absence is exactly what F4 (no aggregate recovery) needs: the Council of Economic Advisers calculated that elevated absence could account for roughly 27% of the 2019–2022 grade 4 mathematics decline and 45% of the reading decline (Council of Economic Advisers, 2023), and Dee (2024) and the Education Recovery Scorecard identify it as a first-order obstacle to recovery. Absence is concentrated among low-income students, fitting the post-2022 divergence between the recovering top and the still-falling bottom.

Verdict. *Confirmed as a significant amplifier of phase two and probably the single most important proximate explanation for the failure to recover; explains little of phase one, though NAEP's own attendance item (Section 7.2) shows disengagement beginning to rise by 2017–2019, earlier than administrative data indicate. Absenteeism is partly a mediator—disengagement itself has causes (H1's disruption of attendance norms, H2's pull of screens).*

5.7 H7: Declining reading practice — well-documented, overlapping H2

The collapse in voluntary reading (Figure 9) is among the largest behavioral shifts measured in any NAEP survey: daily reading for fun at age 13 halved between 2012 and 2023, with most of the fall pre-pandemic. Reading achievement requires volume of practice, and the grade 8 reading series—where losses began earliest (F1) and 46% of the decline predates COVID—is where practice effects should show first. As an explanation this is best understood not as a rival to H2 but as its principal *mechanism* for literacy: the proximate cause is less reading; the distal cause is what replaced it. *Verdict: supported as the mechanism for reading's early and continuing decline.*

5.8 H8: Other candidates

Teacher shortages and quality. Vacancies and underqualified staffing are real ($\geq$36,000 vacant positions and $\geq$163,000 filled by underqualified teachers (Nguyen et al., 2024)) but became acute mainly after 2020 and are concentrated in special education, STEM, and high-poverty schools; teacher-preparation enrollment fell through the 2010s. Plausibly a modest contributor to phase two and to bottom-distribution stagnation; the timing fits phase one poorly. *Verdict: minor contributor.*

Lowered expectations and grade inflation. High-school GPAs rose from 3.17 (2010) to 3.36 (2021) while ACT scores fell (Sanchez and Moore, 2022), and course rigor and homework time show parallel softening. This evidence indicates that *signals* of learning became disconnected from learning, masking the decline from parents and muting corrective pressure—an enabling condition more than a first cause. *Verdict: contributing background condition; direction of causality unclear.*

6 Sharper Tests: Cohorts, Sectors, and Policy Variation

The verdicts above lean on timing and incidence. This section reports three tests designed to discriminate between the two leading phase-one hypotheses (H2 digital media vs. H3 accountability), each exploiting a dimension of the data the candidate explanations treat differently.

6.1 Cohort vs. period: the losses accrue during adolescence

If successive birth cohorts were arriving at school weaker—because of early-childhood environment, poverty, or school quality in the early grades—declines should appear at grade 4 first and follow cohorts upward. If instead something in the post-2012 environment harms whoever is an adolescent at the time, grade 8 should fall regardless of how cohorts looked at grade 4. The data can separate these because the same birth cohort is observed at grade 4 and, four years later, at grade 8.

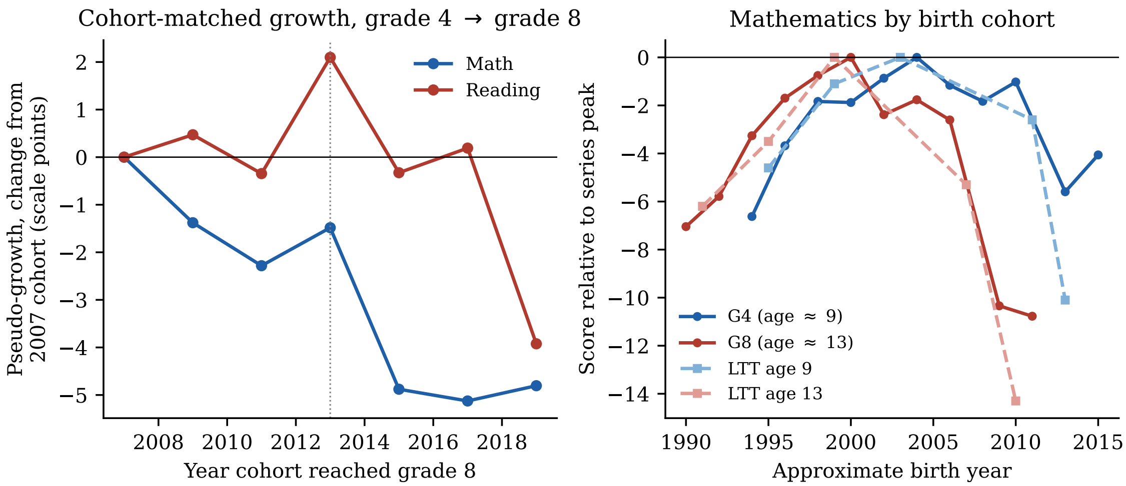

Figure 10: Cohort vs. period. Left: cohort-matched pseudo-growth (grade 8 mean minus the same cohort's grade 4 mean four years earlier), relative to the 2007 value. Right: mathematics scores by approximate birth year for four age groups, each relative to its series peak. Data: NAEP Data Service API; LTT.

Define cohort-matched pseudo-growth as the grade 8 average in year t minus the grade 4 average in year $t-4$. Figure 10 (left) shows this quantity fell sharply for cohorts reaching grade 8 after 2013: mathematics pseudo-growth dropped from 44–46 points (2007–2013 cohorts) to about 41.5 points (2015–2019), and reading from 44–47 to 40.6 by 2019. Decomposing the 2013–2019 grade 8 decline: in mathematics, $-2.6 = +0.7$ (higher grade 4 entry scores of the later cohorts) -3.3 (less growth between grades 4 and 8); in reading, $-4.4 = +1.6 - 6.0$. *The cohorts that drove the pre-pandemic grade 8 decline arrived at grade 4 at record levels and then lost ground during the middle-school years.* The right panel shows the same fact by birth year: the identical cohorts that look healthy in the age-9/grade-4 series collapse in the age-13/grade-8 series once their adolescence falls after roughly 2012.

This is a period effect concentrated in adolescence, not a cohort effect. It is unfavorable to explanations rooted in early childhood or school inputs broadly (which would depress grade 4 entry too), and it matches the exposure profile of smartphones and social media, which saturate ages 11–14 far more than ages 5–9. It is compatible with the accountability hypothesis only in the version where pressure mattered most in the middle grades.

6.2 Public vs. Catholic schools: the decline is sector-neutral

Federal test-based accountability never applied to private schools, while the digital environment engulfed all adolescents. NAEP reports Catholic-school results in every assessment year (overall private results are suppressed in several recent years for participation reasons), giving a usable untreated sector.[2]

Figure 11 shows changes since 2013. Before the pandemic, Catholic schools declined alongside public schools: in grade 8 reading they fell *more* than public schools (means: −7.8 vs. −4.0,

[2] Catholic-school samples are far smaller than public ones, so their estimates are noisier, and Catholic enrollment is selected; level differences are uninformative. The comparison of *trends* is the test.

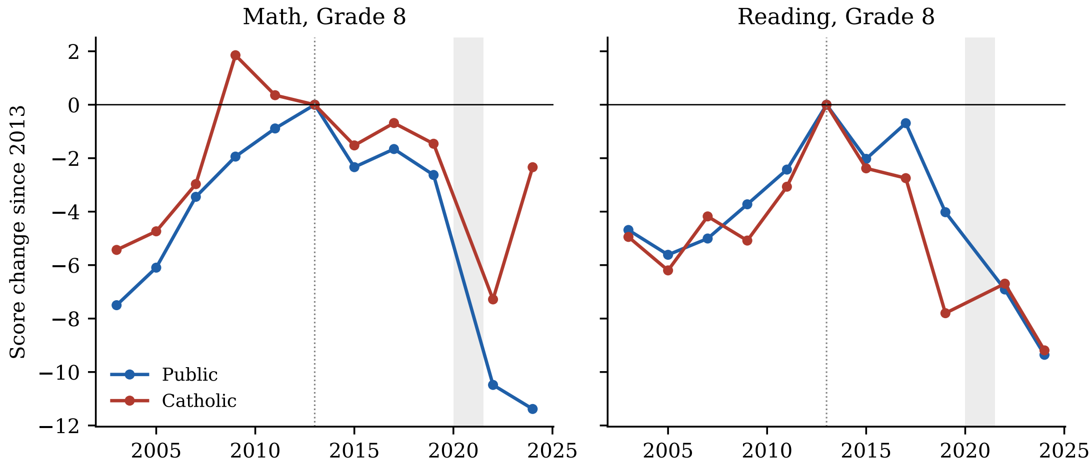

Figure 11: Average scores by school sector, relative to 2013. Data: NAEP Data Service API (SCHTYPE), national.

2013–2019; 10th percentile: −8.7 vs. −10.2); in grade 8 mathematics they fell somewhat less (−1.5 vs. −2.6) but in the same direction, including at the 10th percentile (−3.5 vs. −6.7). A pure accountability story predicts flat private-sector trends; the data instead show the phase-one decline operating inside schools that NCLB and its repeal never touched. (Grade 4 is the partial exception—Catholic grade 4 scores were flat while public grade 4 bottom-decile scores fell—leaving room for an accountability contribution in the early grades.) After 2019 the sectors diverge sharply: by 2024 Catholic grade 8 mathematics had recovered to within 2.3 points of its 2013 level while public schools were 11.4 points below, consistent with private schools' shorter closures and lower post-pandemic absenteeism.

6.3 NCLB waiver timing: no state-level accountability effect

The accountability hypothesis has a directly testable state-level implication: bottom-decile scores should begin falling when a state's accountability pressure was actually released. States received ESEA flexibility waivers at different times—eleven in early 2012, twenty-three more (plus DC) later in 2012, eight in 2013–2014—and seven states (CA, IA, MT, NE, ND, VT, WY) never received one and remained formally under NCLB until ESSA took effect in 2017–18.[3]

Figure 12 shows the test using the state-percentile panel (2003–2019, ending before the pandemic). Left: 10th-percentile grade 8 mathematics scores in early-waiver, late-waiver, and never-waiver states move essentially in lockstep—never-waiver states' bottom decile fell as much as waiver states' after 2012. Right: an event-study regression (early-waiver vs. never-waiver states, all four grade–subject cells pooled in within-cell SD units, state and year fixed effects, clustered by state) finds post-2012 coefficients near zero. A two-way fixed-effects estimate of the post-waiver effect on 10th-percentile scores is +0.09 SD with $p = 0.50$ pooled, and is small and insignificant in every cell separately; effects at the 25th and 90th percentiles are likewise null.

Two caveats temper the inference. The never-waiver group is small (statistical power is

[3]Approval dates compiled from the Department of Education's per-state ESEA flexibility pages, CRS Report R42328, and EdWeek's state-by-state tracking. Washington's waiver was revoked in April 2014; it is dropped from the panel after 2013.

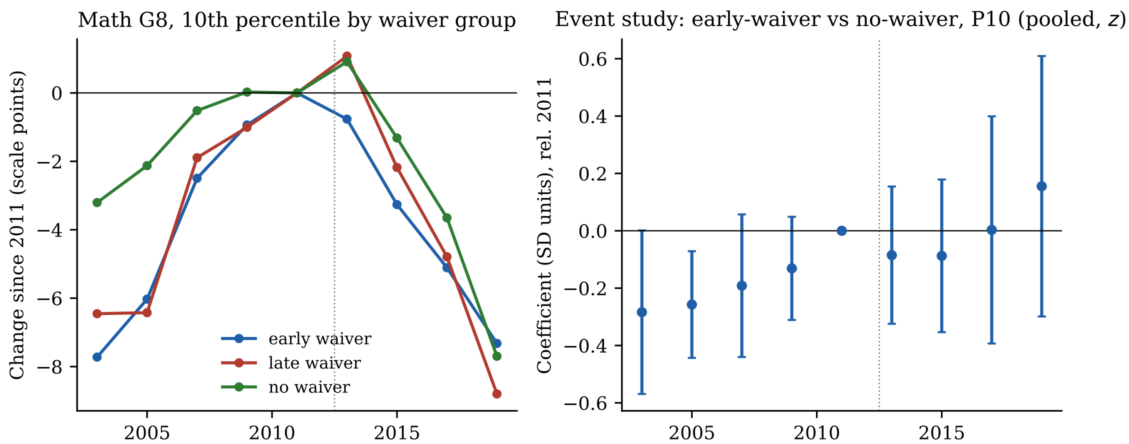

Figure 12: Waiver-timing tests, pre-COVID panel (2003–2019). Left: mean 10th-percentile grade 8 mathematics score by waiver group, relative to 2011. Right: event-study coefficients (early-waiver × year, base 2011, never-waiver comparison), all four grade–subject cells pooled in SD units; bars are 95% confidence intervals, clustered by state. Data: NAEP Data Service API; waiver dates from ED/CRS/EdWeek.

limited, though the point estimates are wrong-signed for H3, not just imprecise), and a waiver is only one notch of deregulation—if the operative change was a national shift in expectations and stakes culminating in ESSA, it would hit all states simultaneously and be absorbed in year effects, invisible to this design. What the test does rule out is the strong form of the hypothesis in which the formal release from NCLB consequences drove a state's low performers downward.

6.4 What the three tests jointly imply

The three tests were designed so the two leading phase-one hypotheses predict different outcomes, and all three landed on the same side. The pre-pandemic decline (i) strikes whoever is an adolescent after ∼2012 regardless of how the cohort looked at age 9, (ii) appears inside schools accountability policy never governed, and (iii) shows no relationship to when states were actually released from NCLB. Each fact is what the digital-media hypothesis predicts and what the school-policy hypothesis does not. The accountability retreat survives only in a weaker form—a national climate change contributing through channels common to all states, with its clearest remaining footprint at the public-school grade 4 bottom decile—while the case that the phase-one decline reflects a change in adolescent life outside the school building is now supported by timing, geography, sector, cohort structure, international synchrony, and the collapse of voluntary reading simultaneously.

7 Mechanism Checks: Schooling Mode and Attendance

Two further analyses probe the phase-two mechanisms directly: merging state NAEP changes with measured pandemic schooling mode, and exploiting NAEP's own student-reported attendance data.

7.1 Schooling mode: the district-level effect mostly washes out between states

Using the COVID-19 School Data Hub's district learning-model records aggregated to enrollment-weighted state shares (47 states plus DC; Iowa, Montana, and Oklahoma are uncovered), states' 2019–2022 NAEP changes can be regressed on the share of the 2020–21 school year spent fully virtual. The relationship is right-signed but weak in mathematics (-0.33 points per 10 percentage points virtual at grade 4, $p = 0.13$; -0.18 at grade 8, $p = 0.35$), absent in reading (grade 8 is wrong-signed), and essentially zero pooled across the four series (-0.04 points per 10pp, $p = 0.80$, clustered by state); Figure 13 (left) shows the grade 4 mathematics scatter. Between-state variation in virtual share explains almost none of the between-state variation in score declines.

This is not a refutation of the closure literature—the district-level studies (Goldhaber et al., 2023; Jack et al., 2023) exploit far larger within-state treatment variation with apter controls, and state aggregation compresses both the treatment (few states were mostly virtual on average) and the outcome (state NAEP changes carry sampling errors of roughly a point) — but it is a useful calibration: *schooling mode explains the timing of the national 2019–2022 drop better than it explains which states fell most.* Factors that varied little between states (the disruption itself, the absenteeism surge, whatever was already eroding the bottom) dominate the state cross-section.

7.2 Attendance: a pre-pandemic disengagement trend, visible inside NAEP

NAEP asks students how many days of school they missed in the month before the assessment, providing an attendance measure tied directly to scores and consistent since 2002. Three facts emerge (Figure 14). First, the share reporting three or more missed days, flat near 19–20% from 2003 through 2013, *began rising before the pandemic*—to 24% at grade 4 by 2019—then jumped to 32–35% in 2022 and remained near 30% in 2024, mirroring the administrative chronic-absenteeism record (Malkus, 2025) and revealing a disengagement pre-trend that the administrative data (flat at ∼15% through 2019) understate. Second, the score penalty associated with absence widened sharply *before* COVID: the gap between students missing no days and those missing more than ten grew from 29 to 42 points in grade 8 reading between 2013 and 2019—absence became more academically costly, or increasingly selected the most disengaged students, exactly when the bottom of the distribution was collapsing. Third, a Kitagawa decomposition splits each mean change into the part from students shifting into higher-absence categories and the part from score declines within categories: the absence shift accounts for roughly 25–42% of the 2019–2024 declines (e.g., 42% in grade 4 mathematics, 25% in grade 8 mathematics), closely bracketing the Council of Economic Advisers' estimate, but for only 7–19% of the pre-pandemic grade 8 declines—phase one was not an attendance phenomenon.

The decomposition is associational (absence proxies broader disengagement), but the pattern coheres with everything above: phase one operated through something other than attendance; phase two converted a disruption into a persistent attendance and engagement problem that the recovery has not closed.

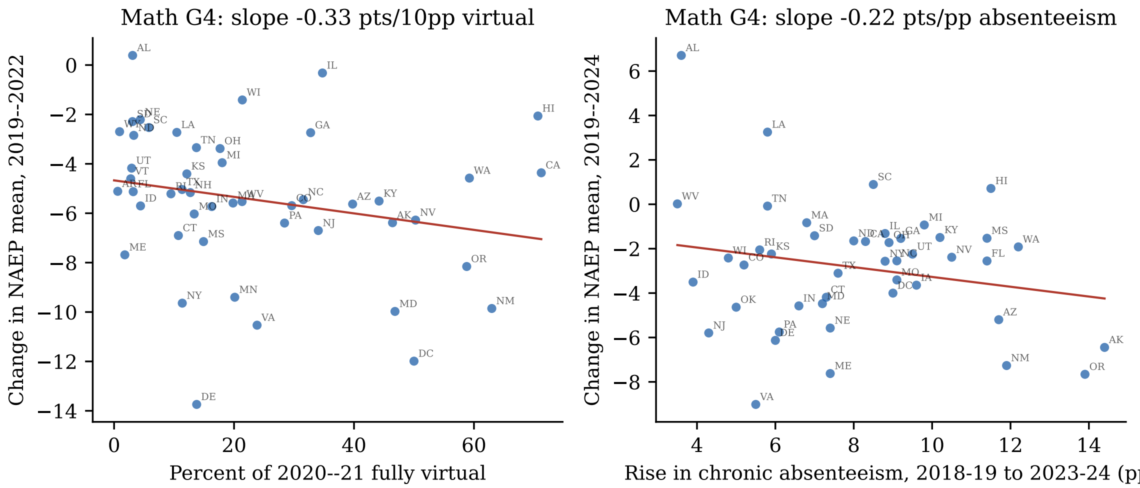

Figure 13: State dose-response. Left: change in grade 4 mathematics, 2019–2022, against the enrollment-weighted share of the 2020–21 school year spent fully virtual (COVID-19 School Data Hub). Right: net change 2019–2024 against the rise in chronic absenteeism, 2018-19 to 2023-24 (FutureEd state tracker).

Table 2: Hypothesis scorecard against the facts of Section 3.

Hypothesis	F1 Timing	F2 Bottom-heavy	F3 COVID drop	F4 No recovery	F5/F7 Within-group	F6 Intern'l
H1 Pandemic	×	∼	✓	∼	✓	∼
H2 Digital media	✓	∼	×	✓	✓	✓
H3 Accountability	✓	✓	×	∼	✓	×
H4 Funding cuts	∼	×	×	×	∼	×
H5 Demographics	×	×	×	×	×	×
H6 Absenteeism	×	✓	✓	✓	✓	∼
H7 Reading practice	✓	∼	×	✓	✓	✓
H8 Teachers/inflation	∼	∼	×	∼	∼	×

✓ = consistent; ∼ = partially consistent; × = inconsistent or silent.

8 Synthesis: A Two-Phase Decline With Different Causes

The evidence is not consistent with any single-cause account, but it is well organized by a two-phase model (Table 2):

Phase one (2013–2019): erosion at the bottom. Slow national decline, concentrated almost entirely among the lowest-performing students, present in most states and many countries, alongside collapsing voluntary reading. Two hypotheses fit the basic facts: the accountability retreat (which predicts the bottom-concentration and the U.S. policy timing, and is corroborated in mirror image by Mississippi) and digital-media displacement (which predicts the international synchrony and the reading-practice collapse). The discriminating tests of Section 6 break the tie in favor of digital media: the losses accrue during adolescence to cohorts that arrived at grade 4 at record levels, they appear equally inside Catholic schools that accountability never governed, and they bear no relationship to when individual states were released from NCLB. The accountability retreat retains a plausible supporting role—as a national climate shift, and

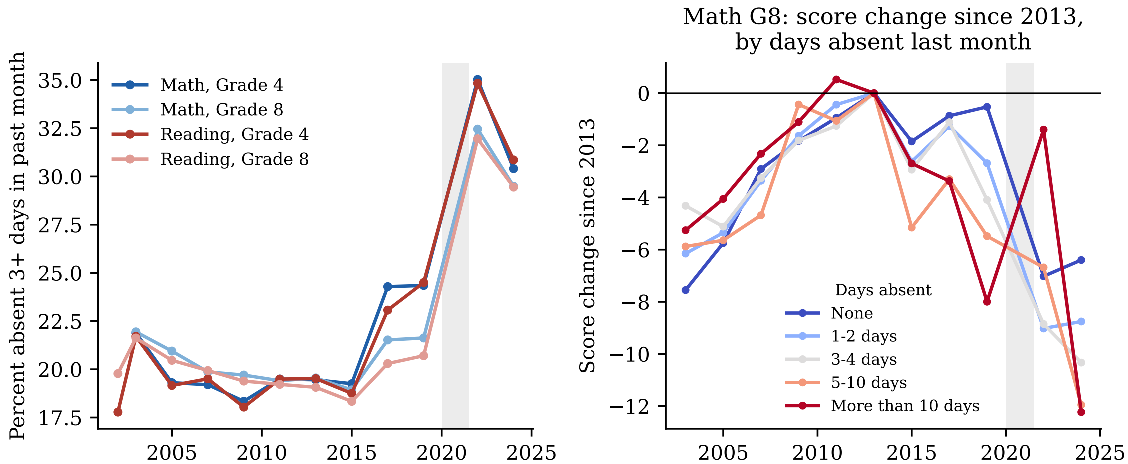

Figure 14: Student-reported absence inside NAEP (B018101). Left: share reporting 3+ missed days in the month before the assessment. Right: grade 8 mathematics score changes since 2013 by absence category. Data: NAEP Data Service API.

at the public-school grade 4 bottom decile, where the sector contrast does point its way—but the strong version in which deregulation drove the floor down is rejected by its own state-level test. Funding, demographics, and teacher supply fail on timing, distribution, or magnitude.

Phase two (2019–2024): shock without recovery. The pandemic caused the large 2019–2022 drop—the schooling-mode dose-response evidence is decisive—and hit the whole distribution. What demands explanation is the aftermath: four-plus years on, aggregate recovery is negligible, reading is still falling, and the bottom decile continues to sink while the top rebounds. Persistently doubled chronic absenteeism is the leading proximate cause, with the phase-one forces (weak accountability pressure, screen displacement) plausibly explaining why disengagement has not self-corrected. On the magnitudes in Table 1, phase one accounts for roughly 20–45% of the total 2013–2024 decline depending on the series, the pandemic window for roughly 30–70%, and the post-2022 period for the remainder—meaning that even a complete reversal of pandemic losses would leave the United States well below its 2013 peak.

Three implications follow. First, "COVID recovery" framing understates the problem: grade 8 reading was already down 4.4 points before the pandemic, and the forces behind that slide are still operating. Second, because losses are concentrated at the bottom, average-score targets understate the equity emergency; the 90–10 gap is the widest ever measured. Third, with the policy-side explanations weakened by the tests of Section 6, the empirical frontier shifts to the digital-media mechanism, where the policy-relevant variation is only now emerging: several countries and many U.S. states have adopted school phone bans since 2023, and their staggered adoption against the next NAEP and PISA waves offers exactly the kind of test that waiver timing provided here. Mississippi's literacy reform remains the strongest evidence that aggressive instructional policy can buck the trend even if policy retreat did not cause it.

9 Limitations

This analysis is observational. The hypothesis tests rest on timing, distributional incidence, and cross-sectional comparisons, not experimental variation; several causes plausibly interact (absenteeism is partly downstream of screens and of weakened school engagement), so the phase shares above are accounting decompositions, not causal attributions. NAEP percentile statistics for the oldest years use no-accommodation samples; the NSLP-eligibility series ends in 2022 and is affected by community-eligibility expansion; PISA OECD averages shift membership across cycles; and the LTT format changed in 2004 (results are bridged). Excluded-student rates and assessment-participation changes could in principle affect trends, though NCES validity checks have not found effects of the magnitude observed. The Section 6 tests carry their own caveats: the never-waiver comparison group is seven states (limited power, and visible pre-2011 trend differences), waivers capture only the formal step of deregulation, and Catholic-school samples are small and selected, so only their trends—not levels—are informative. The Section 7 state regressions are aggregation-limited (state-level treatment variance is a fraction of district-level variance, and three states lack schooling-mode data), NAEP's absence item covers only the month before the assessment, and the Kitagawa decomposition treats absence as exogenous when it partly proxies disengagement. Finally, the digital-media evidence, while consistent across timing, international, survey, cohort, and sector data, remains associational at the individual level.

10 Conclusion

The decline in American student achievement is real, large, and verifiable from primary data: roughly one grade level of learning lost at grade 8 since 2013, the lowest grade 8 reading scores ever recorded, and the widest achievement gaps in NAEP's history. It happened in two phases with different causes. A pre-pandemic erosion, nearly invisible in averages because it was confined to the bottom of the distribution, is best explained by the saturation of adolescent life by smartphones and digital media after 2012: the losses strike whoever is an adolescent after that date (not particular cohorts), appear identically in Catholic schools that accountability policy never governed, show no relationship to when states were released from NCLB, recur across the rich world, and coincide with a halving of voluntary reading. The retreat from test-based accountability likely contributed at the margins—most visibly at the public-school grade 4 bottom decile—but failed the direct tests this report could put to it. The pandemic then imposed the largest schooling shock in modern history, whose effects, far from fading, have been locked in by chronic absenteeism that remains half again its pre-pandemic level. Funding cuts, demographic change, and teacher shortages are, on the evidence, second-order. The policy conclusion is uncomfortable but clear: returning to 2019 practices would only return the country to a trajectory that was already pointing down.

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